MIT IT Project Workshop

Visualisation with Geospatial Data

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media design hochschule

Autumn 2025

Visualising Quantitative Data in Maps

User requirements asking for displaying spatial relations in the analytical application

Explicit requirements for spatial relations in the Consolidated User Requirements

Spatial	S-01	Spatial	District region (BZR) level mapping	Lisa, Markus
	S-02	Spatial	Color-coded choropleth maps for hotspots	Lisa, Markus
	S-03	Spatial	Prevalence rates per 100k inhabitants (CRITICAL)	Markus Keller
	S-04	Spatial	Top 10 BZR rankings by various metrics	Markus Keller
	S-05	Spatial	Top 20 BZR detailed rankings	Markus, Tanja
	S-06	Spatial	Sortable comparative tables across regions	Markus, Tanja

User Research has shown that all user groups need to show measures in relation to spatial distribution in Berlin from high levels to low levels of aggregation.

Spatial Data Relations

Spatial visualisations are no longer charts in the traditional sense, but a combination of graphics plotted as layers on top of a geographic map.

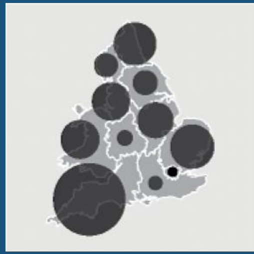
Dot Density Map

Showing where certain things/events are located and at what density allows us to visualise the local distribution of clusters in contrast to dispersed patterns.

Choropleth Maps

Colouring of regional areas according to the values of the variable of interest.
Values are grouped into categories.

Proportional Symbol Map



The size of a single symbol encodes an amount or value (typically values grouped into categories).

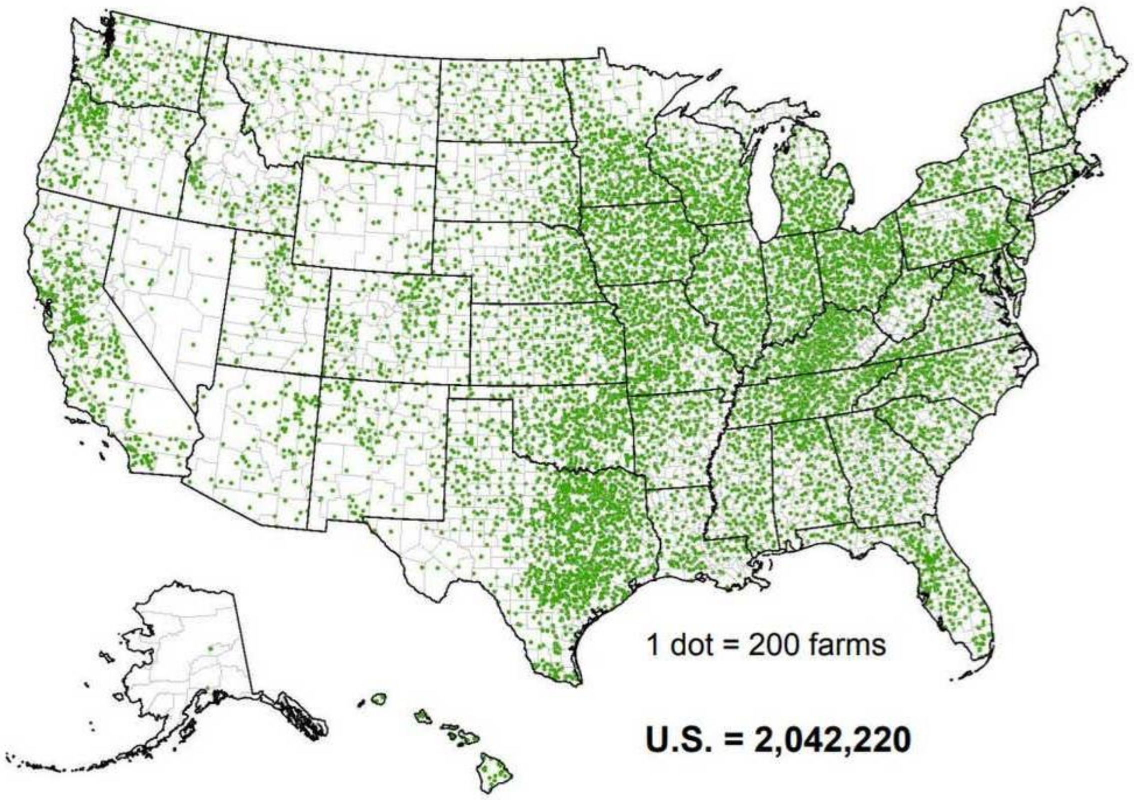
Dot Density Map

Names:
Dot Density Map, Dot Distribution Map

A dot density map is a type of map that uses dot symbols to show the presence of a feature or phenomenon.

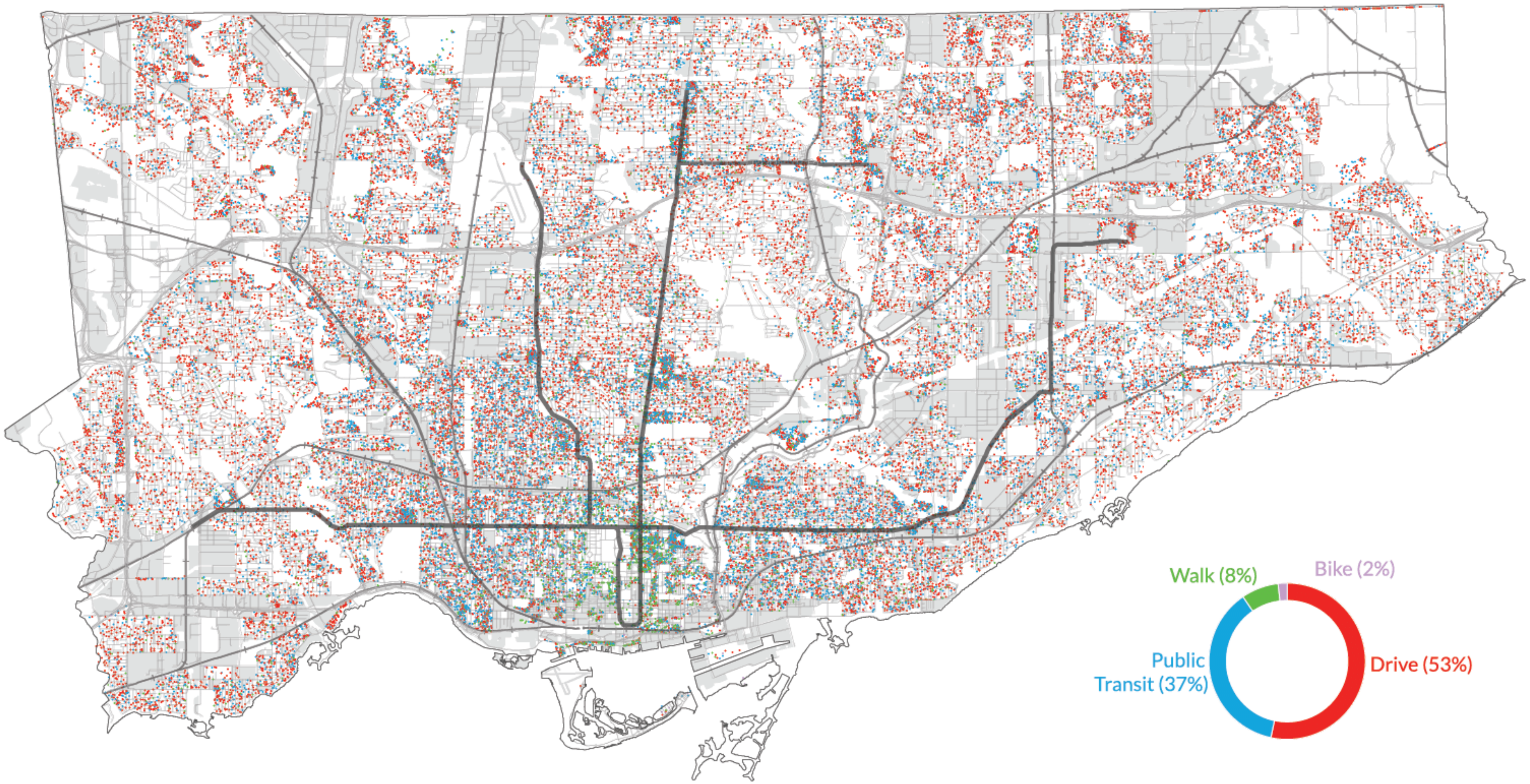
https://en.wikipedia.org/wiki/Dot_distribution_map

- Visualisation of where certain things or events are located and at what density
- Shows the local distribution of clusters versus dispersion
- Scaling:
  - One-to-one: 1 dot = 1 item/event
  - One-to-many: 1 dot = xxx items/events (e.g. 1,000 people, 200 farms)



https://eu.wisfarmer.com/story/news/2019/04/17/data-2017-census-paints-telling-picture-agriculture/3490020002/

Dot Density Map Example: Primary Mode of Transportation to Work in Toronto



● each dot represents 20 people

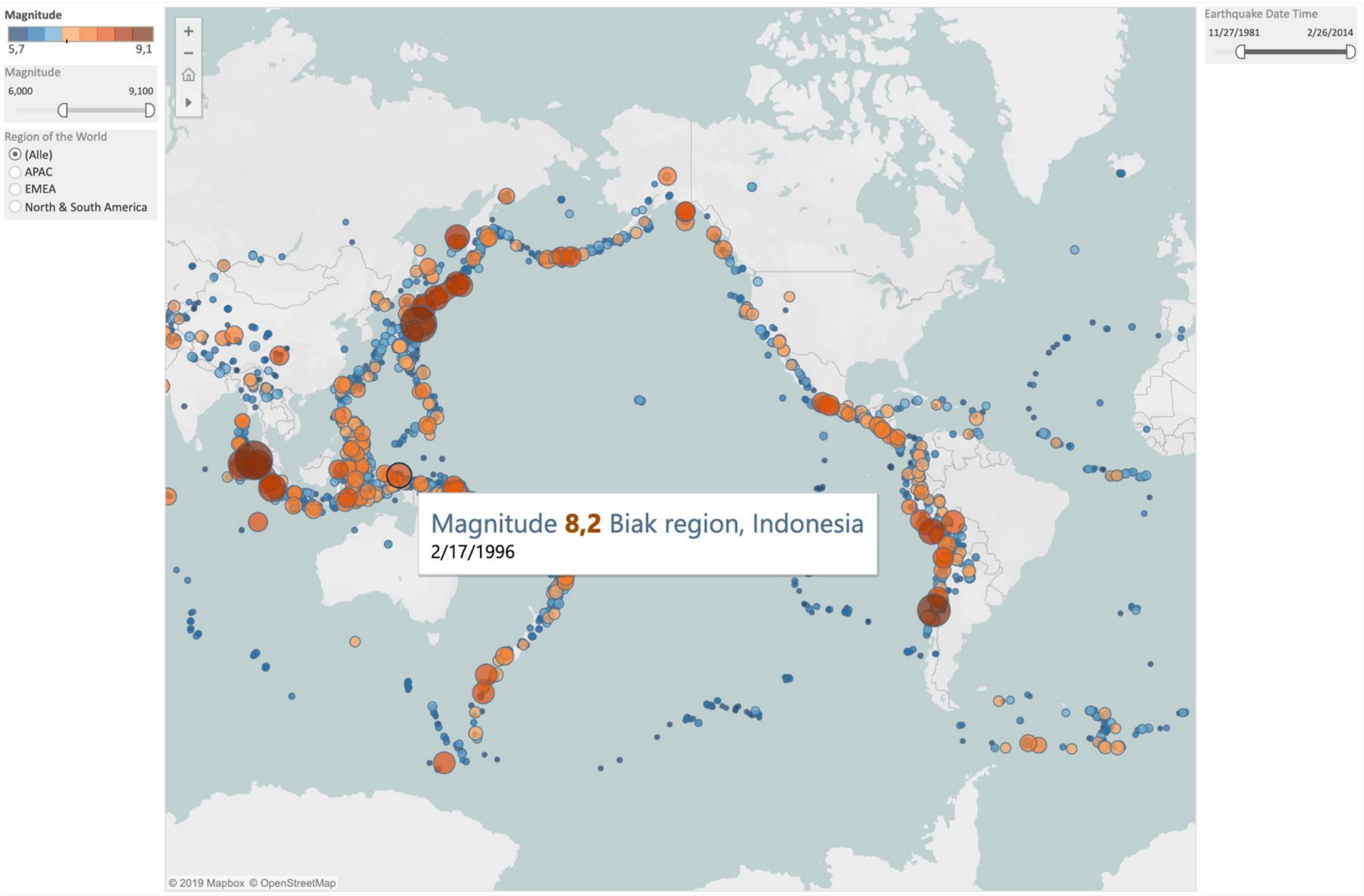
0 5 Km

Jonathan Critchley, 2012

Data Source: DMTI Spatial Inc. CanMap RouteLogistics 2007.3 (City of Toronto): Census [digital resource: vector]. Markham, Ontario: DMTI Spatial Inc., 2007. Retrieved (29, November, 2011).
Statistics Canada. 2006. Census of Canada 2006 Digital Cartographic File (cartographic boundary file, [Dissemination Area]). Retrieved (29, November, 2011).

http://jonathancritchley.ca/todot.html

Proportional Symbol Karte



https://public.tableau.com/app/profile/tableau.docs.team/viz/CreateProportionalSymbolMapsinTableauExampleWorkbook/Map

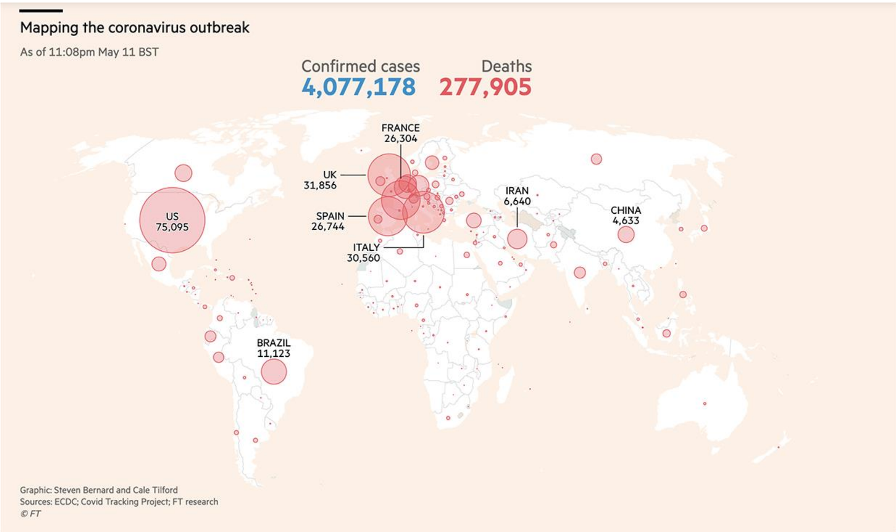
A **proportional symbol map** displays **quantitative values** for locations on a map.

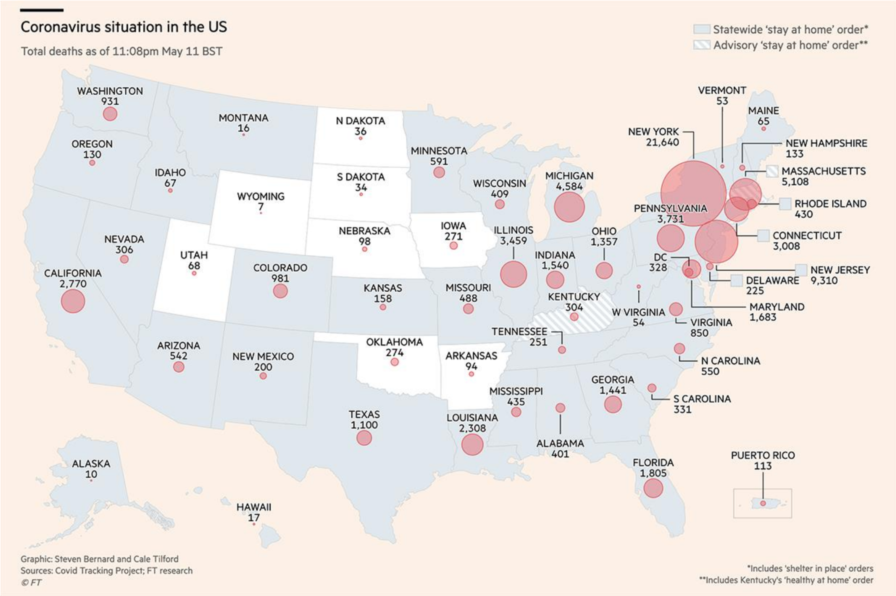
The values are represented by **proportionally scaled shapes** (usually circles) positioned with their centre over a specific location.

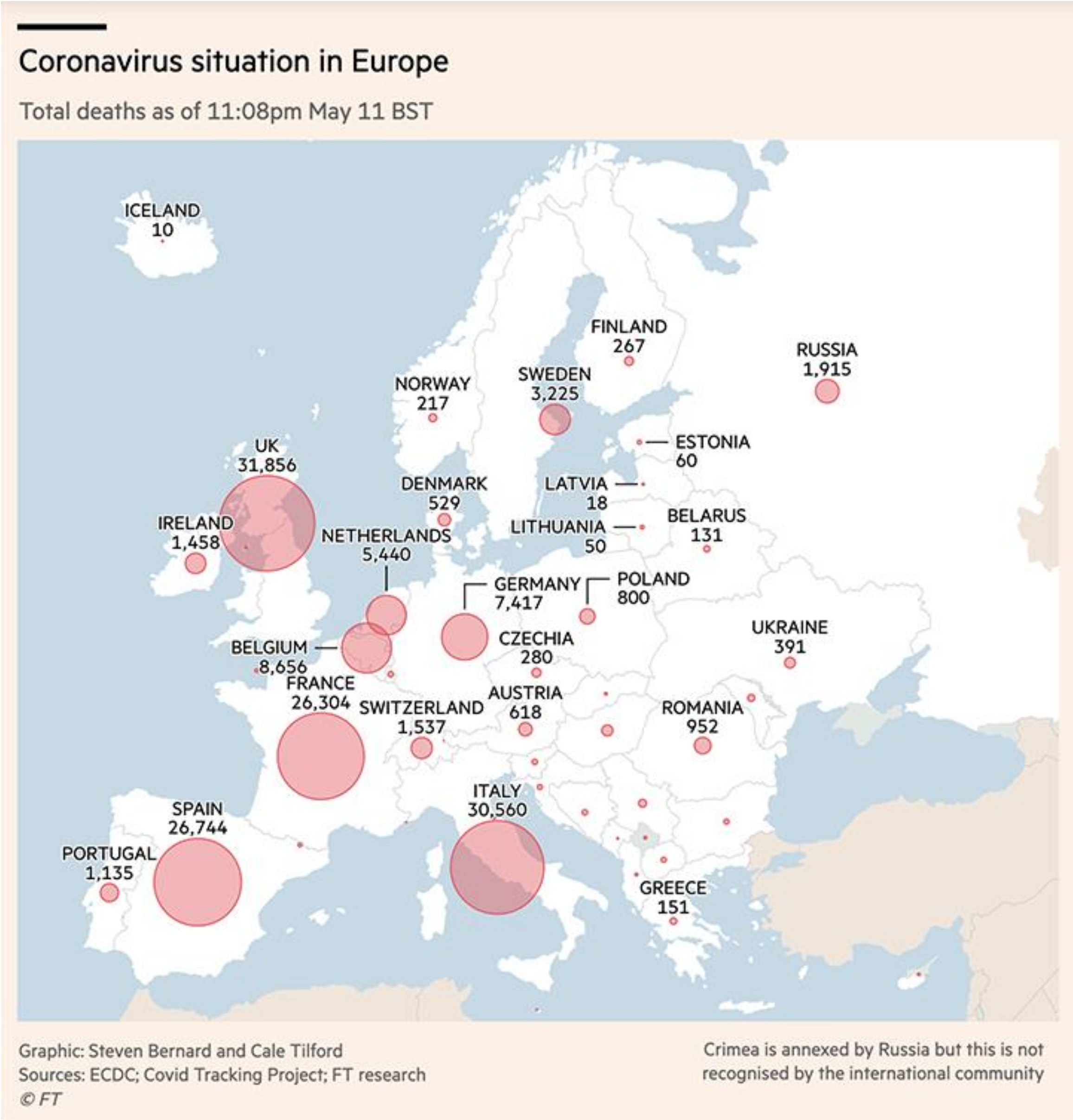
Colour is sometimes used to introduce an additional **categorical distinction**

Kirk: Data Visualisation, p.182

Proportional Symbol Karte: Beispiele aus der Coronavirus Zeit

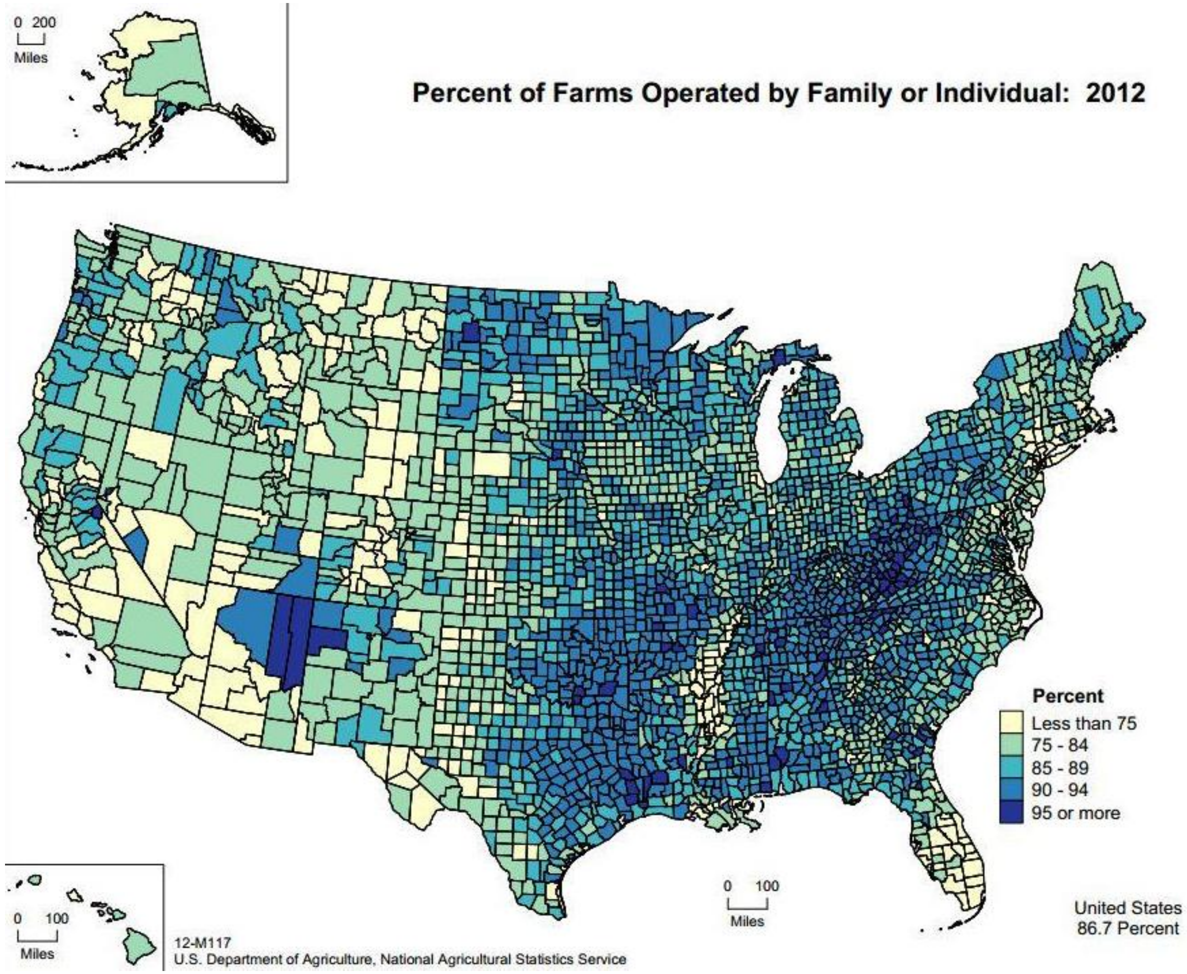






Quelle (Charts nicht mehr vorhanden) https://www.ft.com/content/a26fbf7e-48f8-11ea-aeb3-955839e06441

Choropleth Map



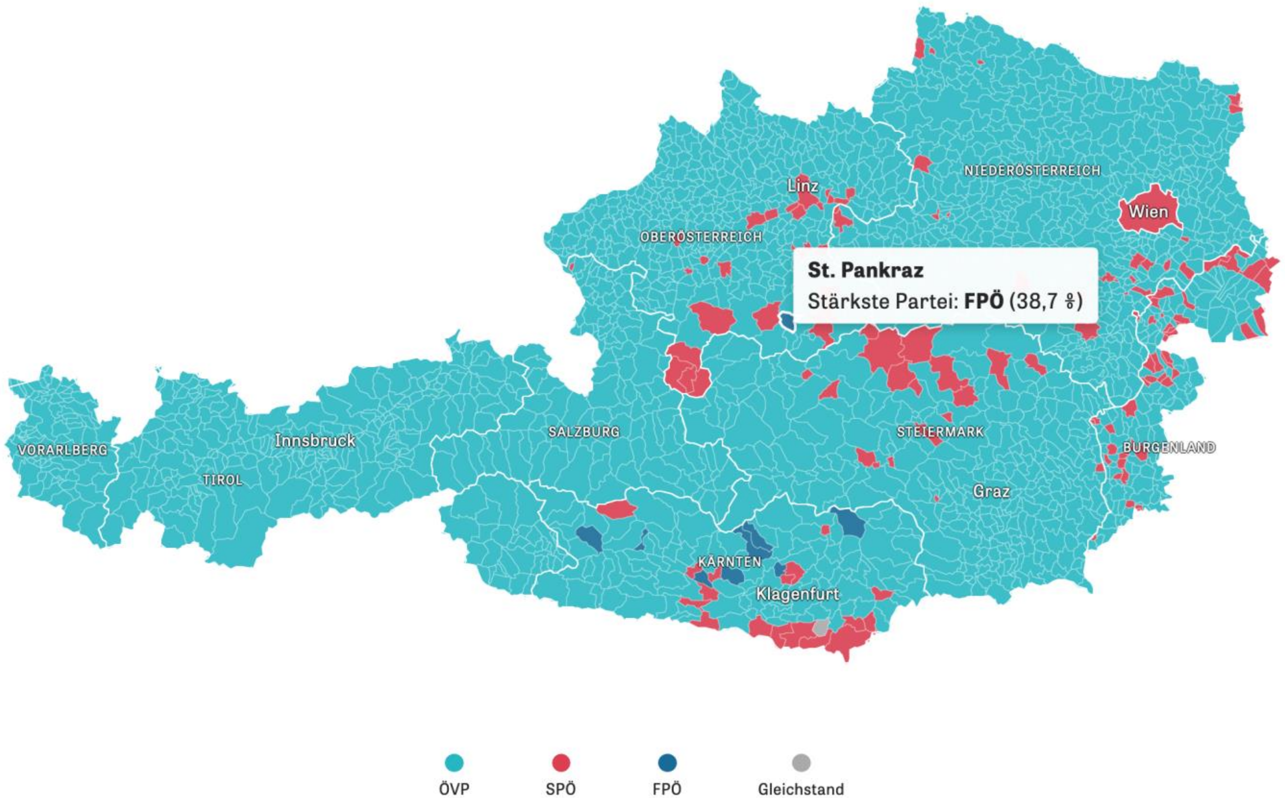
Choropleth Karte

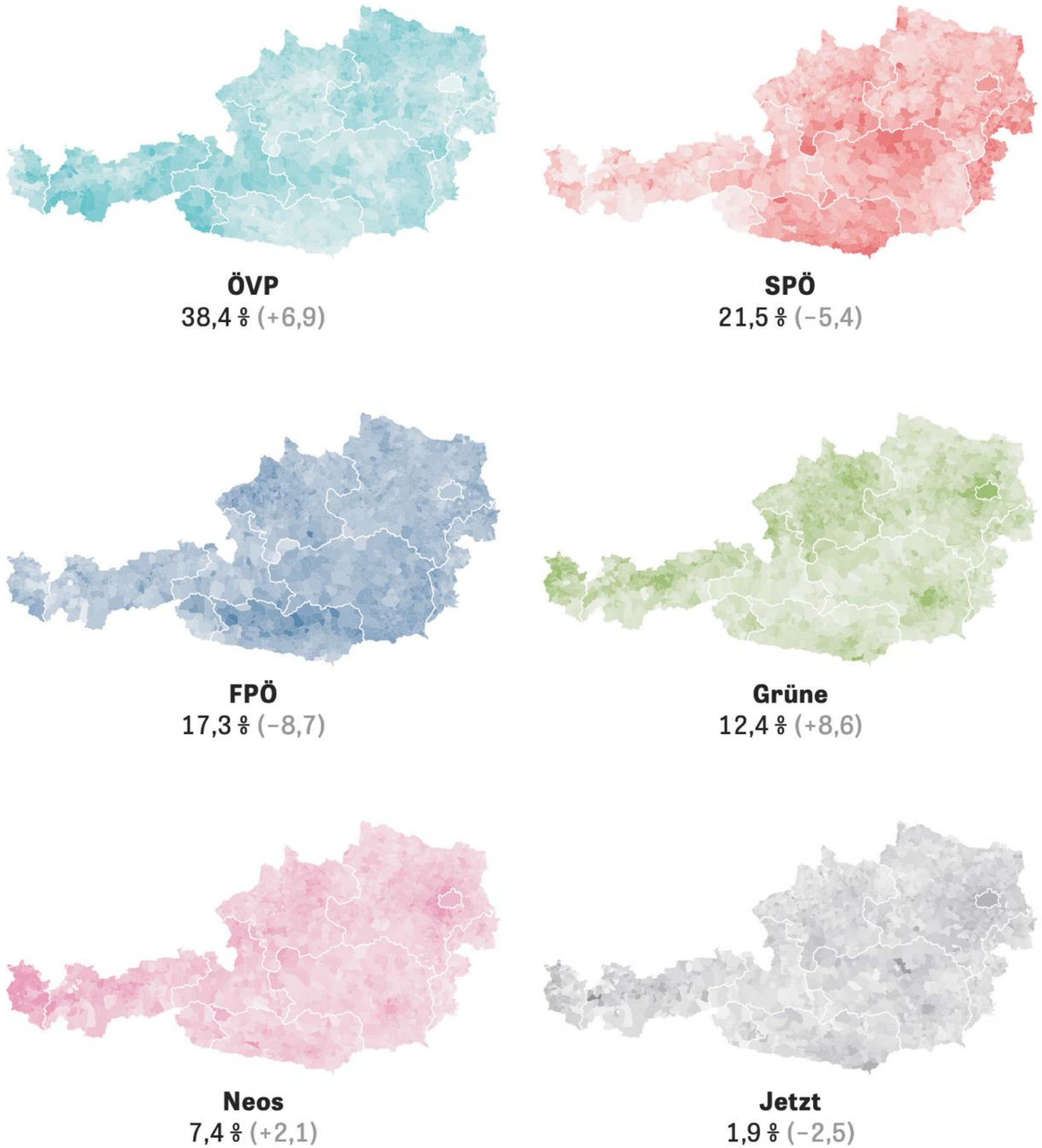
A **choropleth map** is a **thematic map** in which areas are shaded or patterned **in proportion to the measurement of a variable**.

Characteristics

- Typically, some form of **sequential colour scale** is used.
- Sometimes a **diverging colour scale** is applied.
- Additional variables may be encoded using patterns (e.g. hatching, dots, etc.).
- The visual complexity usually requires that value labels are omitted within the map.

Choropleth Map Examples: Elections in Austria 2019





Visualising Quantitative Data in Maps

The NUTS level system

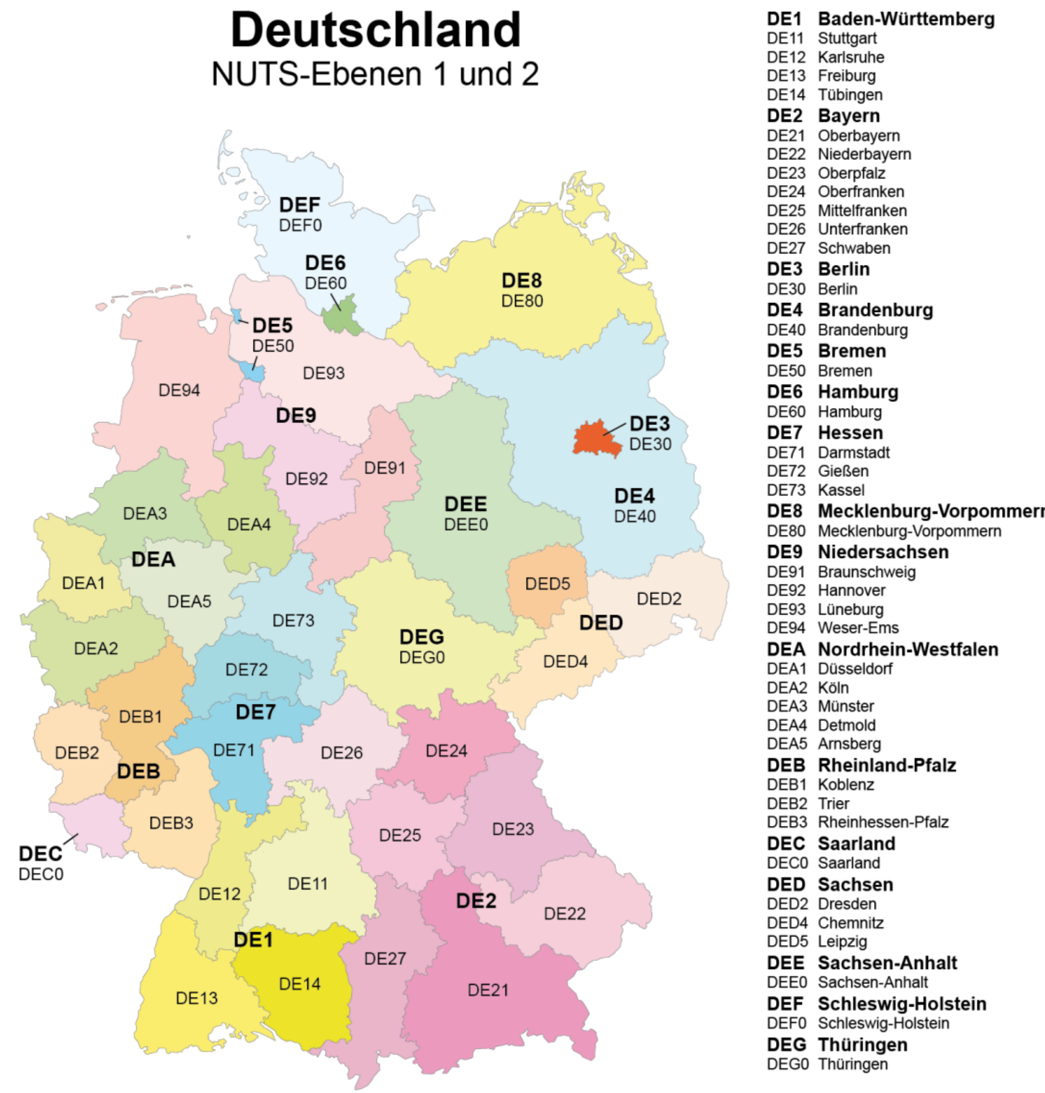
The NUTS level system (Nomenclature of Territorial Units for Statistics) is a hierarchical geographic classification used by the European Union to divide up countries into regions for statistical and analytical purposes.

It has three main levels:
NUTS 1: 16 federal states (Bundesländer)
NUTS 2: 38 regions (administrative districts, city-states, etc.)
NUTS 3: Small regions for specific diagnoses (e.g. districts or counties)

More on the NUTS levels:
https://en.wikipedia.org/wiki/NUTS:DE

Within Tableau, map data is only built in up to NUTS levels 0, 1, and 2.

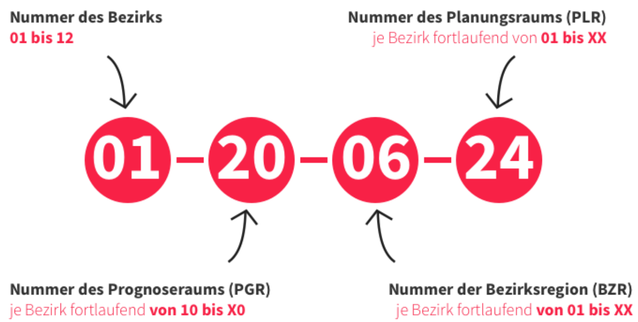
Berlin is considered a single unit at NUTS levels 1, 2, and 3. The inner-city districts are not specified as substructures in either the NUTS classification or in Tableau.
Therefore, for inner-city subdivisions, we need to rely on other regional breakdowns.

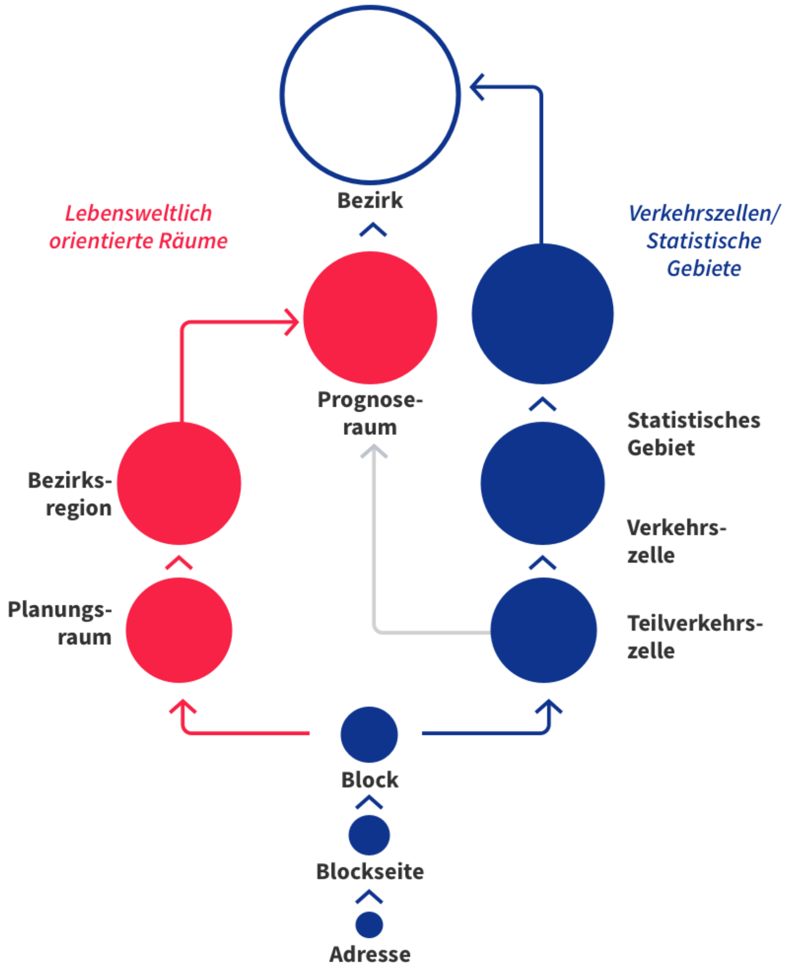


Maximilian Dörrbecker (Chumwa) - Eigenes Werk, CC BY-SA 2.5,
https://commons.wikimedia.org/w/index.php?curid=6779016

LOR – Spatial structure of Berlin used in our data set with fine granularity

- The most important small-area spatial structure in Berlin is the system of Lebensweltlich orientierte Räume (LOR, life-world oriented areas).
- This fine-grained structure is used as a localization and assignment system by local administrations.
- It serves as an organizational tool to support administrative processes and statistical surveys.
- The defined subareas can be clearly delineated from one another and are mutually comparable.





Source: https://www.statistik-berlin-brandenburg.de/meine-region/lebensweltlich-orientierte-raeume-berlin#

The LOR Structure: Berlin's Life-world-oriented spatial structure

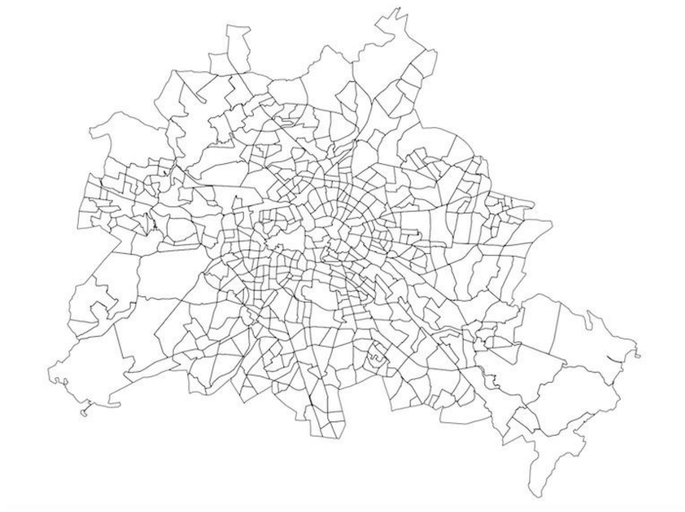
12 Bezirke (districts)

Bounderies of Berlin's 12 districts

143 Bezirksregionen (district regions)

143 district regions as a level of the life-world-oriented spatial structure (LOR), aligned with the administrative organisation of the district offices. On average, each district region comprises about 25,000 inhabitants.

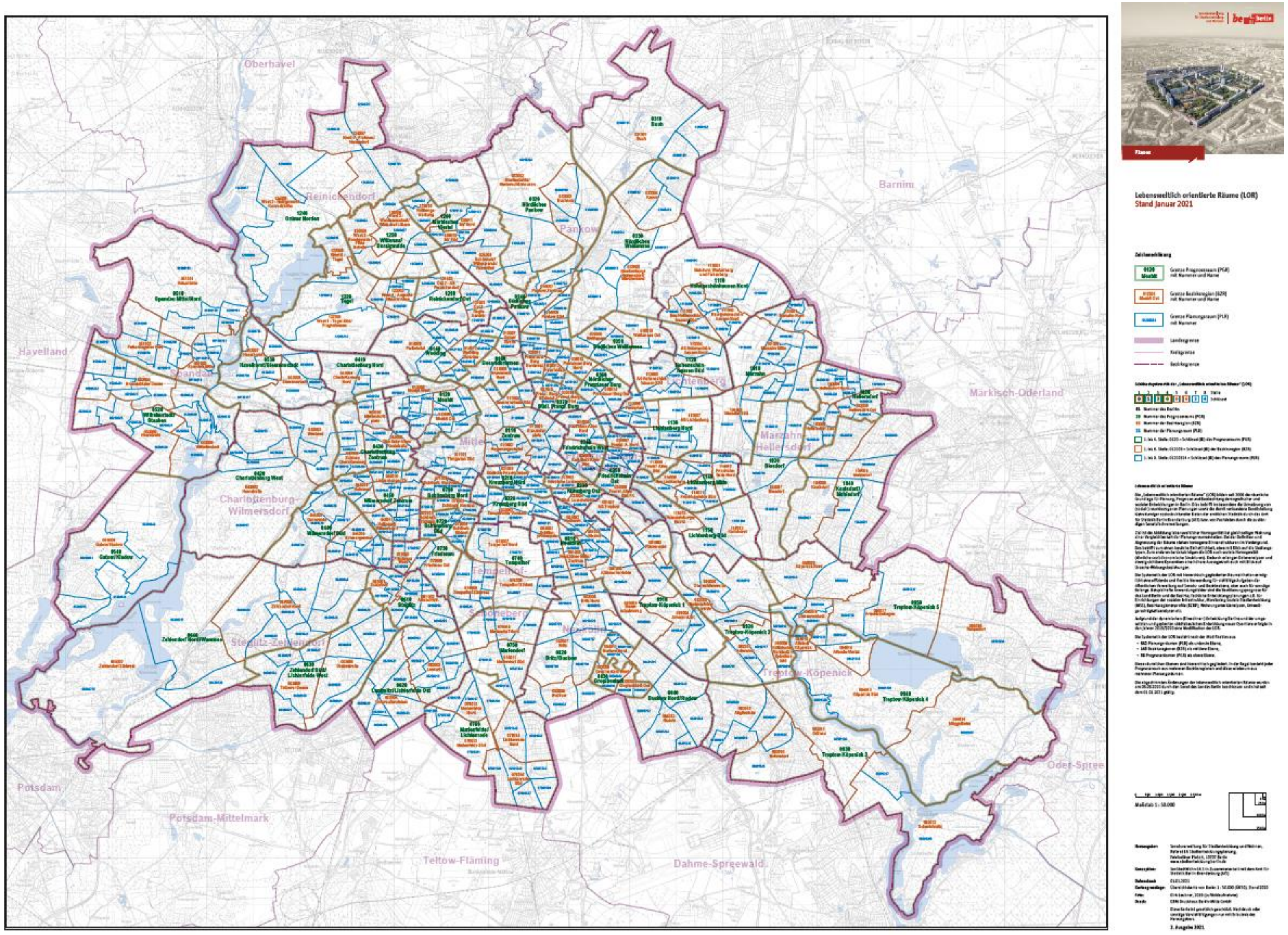
524 Planungsräume



Planning Areas represent the lowest observation and planning level of the life-world-oriented spatial hierarchy (LOR) within the Regional Reference System (RBS). On average, each planning area comprises about 7,500 inhabitants.

Source: Technologiestiftung Berlin (o.D.), Basisdatensätze von Berlin
https://daten.odis-berlin.de/

Map with all LOR-Levels



- The map is displaying all LOR-levels.
- File „Large Map with all LOR Levels.pdf“ is in the Teams repository

Source: https://www.berlin.de/sen/sbw/_assets/stadtdaten/stadtwissen/lebensweltlichorientierte-raeume/lor_2021_kartenseite-2.pdf?ts=1752674586

Excel file ’LOR all Levels.xlsx’

Tab „LOR_2023_Übersicht“

	A	B	C	D	E	F	G	H	I	J
1	BLN_ID	BLN	BEZ_ID	BEZ	PGR_ID	PGR	BZR_ID	BZR	PLR_ID	PLR
2	00	Berlin	01	Mitte	0110	Zentrum	011001	Tiergarten Süd	01100101	Stülerstraße
3	00	Berlin	01	Mitte	0110	Zentrum	011001	Tiergarten Süd	01100102	Großer Tiergarten
4	00	Berlin	01	Mitte	0110	Zentrum	011001	Tiergarten Süd	01100103	Lützowstraße
5	00	Berlin	01	Mitte	0110	Zentrum	011001	Tiergarten Süd	01100104	Körnerstraße
6	00	Berlin	01	Mitte	0110	Zentrum	011002	Regierungsviertel	01100205	Wilhelmstraße
7	00	Berlin	01	Mitte	0110	Zentrum	011002	Regierungsviertel	01100206	Unter den Linden
8	00	Berlin	01	Mitte	0110	Zentrum	011002	Regierungsviertel	01100207	Leipziger Straße
9	00	Berlin	01	Mitte	0110	Zentrum	011003	Alexanderplatz	01100308	Charitéviertel
10	00	Berlin	01	Mitte	0110	Zentrum	011003	Alexanderplatz	01100309	Oranienburger Straße
11	00	Berlin	01	Mitte	0110	Zentrum	011003	Alexanderplatz	01100310	Alexanderplatzviertel
12	00	Berlin	01	Mitte	0110	Zentrum	011003	Alexanderplatz	01100311	Karl-Marx-Allee
13	00	Berlin	01	Mitte	0110	Zentrum	011003	Alexanderplatz	01100312	Heine-Viertel West
14	00	Berlin	01	Mitte	0110	Zentrum	011003	Alexanderplatz	01100313	Heine-Viertel Ost
15	00	Berlin	01	Mitte	0110	Zentrum	011004	Brunnenstraße Süd	01100414	Nordbahnhof
16	00	Berlin	01	Mitte	0110	Zentrum	011004	Brunnenstraße Süd	01100415	Invalidenstraße
17	00	Berlin	01	Mitte	0110	Zentrum	011004	Brunnenstraße Süd	01100416	Arkonaplatz
18	00	Berlin	01	Mitte	0120	Moabit	012005	Moabit West	01200517	Huttenkiez
19	00	Berlin	01	Mitte	0120	Moabit	012005	Moabit West	01200518	Beusselkiez
20	00	Berlin	01	Mitte	0120	Moabit	012005	Moabit West	01200519	Emdener Straße
21	00	Berlin	01	Mitte	0120	Moabit	012005	Moabit West	01200520	Bremer Straße
22	00	Berlin	01	Mitte	0120	Moabit	012005	Moabit West	01200521	Zwinglistraße
23	00	Berlin	01	Mitte	0120	Moabit	012005	Moabit West	01200522	Elberfelder Straße
24	00	Berlin	01	Mitte	0120	Moabit	012006	Moabit Ost	01200623	Stephankiez
25	00	Berlin	01	Mitte	0120	Moabit	012006	Moabit Ost	01200624	Heidestraße
26	00	Berlin	01	Mitte	0120	Moabit	012006	Moabit Ost	01200625	Lübecker Straße
27	00	Berlin	01	Mitte	0120	Moabit	012006	Moabit Ost	01200626	Thomasiusstraße
28	00	Berlin	01	Mitte	0120	Moabit	012006	Moabit Ost	01200627	Zillesiedlung
29	00	Berlin	01	Mitte	0120	Moabit	012006	Moabit Ost	01200628	Lüneburger Straße
30	00	Berlin	01	Mitte	0120	Moabit	012006	Moabit Ost	01200629	Hansaviertel
31	00	Berlin	01	Mitte	0130	Gesundbrunnen	013007	Osloer Straße	01300730	Drontheimer Straße
32	00	Berlin	01	Mitte	0130	Gesundbrunnen	013007	Osloer Straße	01300731	Koloniestraße
33	00	Berlin	01	Mitte	0130	Gesundbrunnen	013007	Osloer Straße	01300732	Soldiner Straße
34	00	Berlin	01	Mitte	0130	Gesundbrunnen	013007	Osloer Straße	01300733	Gesundbrunnen
35	00	Berlin	01	Mitte	0130	Gesundbrunnen	013008	Brunnenstraße Nord	01300834	Brunnenstraße
36	00	Berlin	01	Mitte	0130	Gesundbrunnen	013008	Brunnenstraße Nord	01300835	Humboldthain Süd
37	00	Berlin	01	Mitte	0130	Gesundbrunnen	013008	Brunnenstraße Nord	01300836	Humboldthain Nordwest
38	00	Berlin	01	Mitte	0140	Wedding	014009	Parkviertel	01400937	Afrikanische Straße
39	00	Berlin	01	Mitte	0140	Wedding	014009	Parkviertel	01400938	Kameruner Straße
40	00	Berlin	01	Mitte	0140	Wedding	014009	Parkviertel	01400939	Glasgower Straße

- We will use this table „LOR_2023_Übersicht“ as a lookup table in our application.
- Purpose: We want be able to display to the users the related spatial entities by name and ID of
  - of an particular single biketheft incident
  - of any aggregation level on the time dimension
  - on the spatial levels of
    - district / Bezirk / BZ
    - district region / Bezirksregion / BZR
    - planning area / Planungsraum / PLR

Conversion into a .csv file

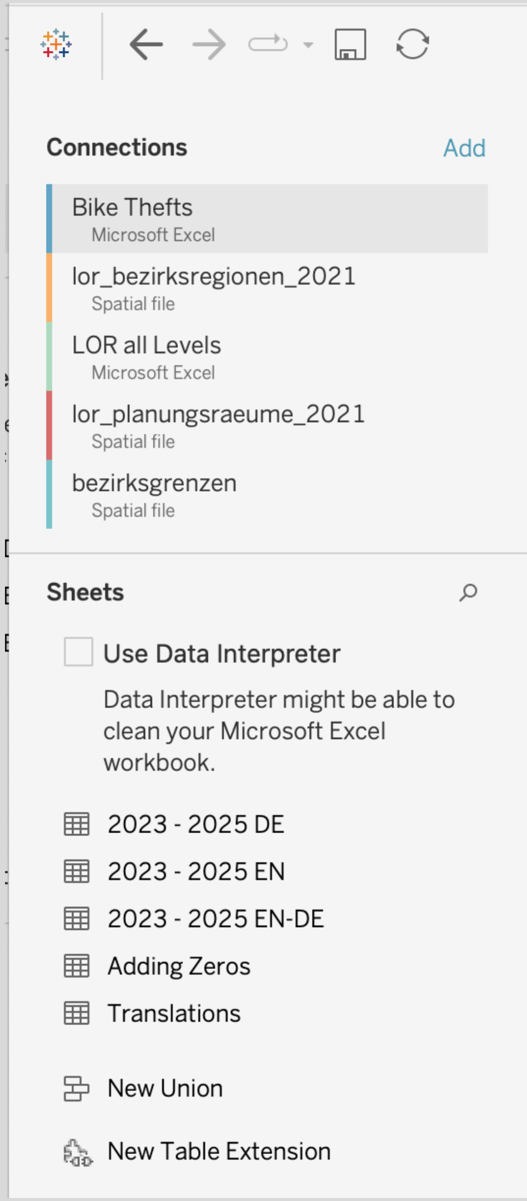
lor_lookup.csv

```
BLN_ID,BLN,BEZ_ID,BEZ_Name,PGR_ID,PGR_Name,BZR_ID,BZR_Name,PLR_ID,PLR_Name
00,Berlin,01,Mitte,0110,Zentrum,011001,Tiergarten Süd,01100101,Stülerstraße
00,Berlin,01,Mitte,0110,Zentrum,011001,Tiergarten Süd,01100102,Großer Tiergarten
00,Berlin,01,Mitte,0110,Zentrum,011001,Tiergarten Süd,01100103,Lützowstraße
00,Berlin,01,Mitte,0110,Zentrum,011001,Tiergarten Süd,01100104,Körnerstraße
00,Berlin,01,Mitte,0110,Zentrum,011002,Regierungsviertel,01100205,Wilhelmstraße
00,Berlin,01,Mitte,0110,Zentrum,011002,Regierungsviertel,01100206,Unter den Linden
00,Berlin,01,Mitte,0110,Zentrum,011002,Regierungsviertel,01100207,Leipziger Straße
00,Berlin,01,Mitte,0110,Zentrum,011003,Alexanderplatz,01100308,Charitéviertel
00,Berlin,01,Mitte,0110,Zentrum,011003,Alexanderplatz,01100309,Oranienburger Straße
00,Berlin,01,Mitte,0110,Zentrum,011003,Alexanderplatz,01100310,Alexanderplatzviertel
00,Berlin,01,Mitte,0110,Zentrum,011003,Alexanderplatz,01100311,Karl-Marx-Allee
00,Berlin,01,Mitte,0110,Zentrum,011003,Alexanderplatz,01100312,Heine-Viertel West
00,Berlin,01,Mitte,0110,Zentrum,011003,Alexanderplatz,01100313,Heine-Viertel Ost
00,Berlin,01,Mitte,0110,Zentrum,011004,Brunnenstraße Süd,01100414,Nordbahnhof
00,Berlin,01,Mitte,0110,Zentrum,011004,Brunnenstraße Süd,01100415,Invalidenstraße
00,Berlin,01,Mitte,0110,Zentrum,011004,Brunnenstraße Süd,01100416,Arkonaplatz
00,Berlin,01,Mitte,0120,Moabit,012005,Moabit West,01200517,Huttenkiez
00,Berlin,01,Mitte,0120,Moabit,012005,Moabit West,01200518,Beusselkiez
00,Berlin,01,Mitte,0120,Moabit,012005,Moabit West,01200519,Emdener Straße
00,Berlin,01,Mitte,0120,Moabit,012005,Moabit West,01200520,Bremer Straße
00,Berlin,01,Mitte,0120,Moabit,012005,Moabit West,01200521,Zwinglistraße
00,Berlin,01,Mitte,0120,Moabit,012005,Moabit West,01200522,Elberfelder Straße
00,Berlin,01,Mitte,0120,Moabit,012006,Moabit Ost,01200623,Stephankiez
00,Berlin,01,Mitte,0120,Moabit,012006,Moabit Ost,01200624,Heidestraße
00,Berlin,01,Mitte,0120,Moabit,012006,Moabit Ost,01200625,Lübecker Straße
00,Berlin,01,Mitte,0120,Moabit,012006,Moabit Ost,01200626,Thomasiusstraße
00,Berlin,01,Mitte,0120,Moabit,012006,Moabit Ost,01200627,Zillesiedlung
00,Berlin,01,Mitte,0120,Moabit,012006,Moabit Ost,01200628,Lüneburger Straße
00,Berlin,01,Mitte,0120,Moabit,012006,Moabit Ost,01200629,Hansaviertel
00,Berlin,01,Mitte,0130,Gesundbrunnen,013007,Osloer Straße,01300730,Drontheimer Straße
00,Berlin,01,Mitte,0130,Gesundbrunnen,013007,Osloer Straße,01300731,Koloniestraße
00,Berlin,01,Mitte,0130,Gesundbrunnen,013007,Osloer Straße,01300732,Soldiner Straße
00,Berlin,01,Mitte,0130,Gesundbrunnen,013007,Osloer Straße,01300733,Gesundbrunnen
00,Berlin,01,Mitte,0130,Gesundbrunnen,013008,Brunnenstraße Nord,01300834,Brunnenstraße
```

- I converted the relevant tab already into .csv file
- You can use for your implementation either .xlsx or .csv file.

Visualising Quantitative Data in Maps

Files needed for visualising bike theft data on a map



Crime File

- Bike Thefts.xlsx

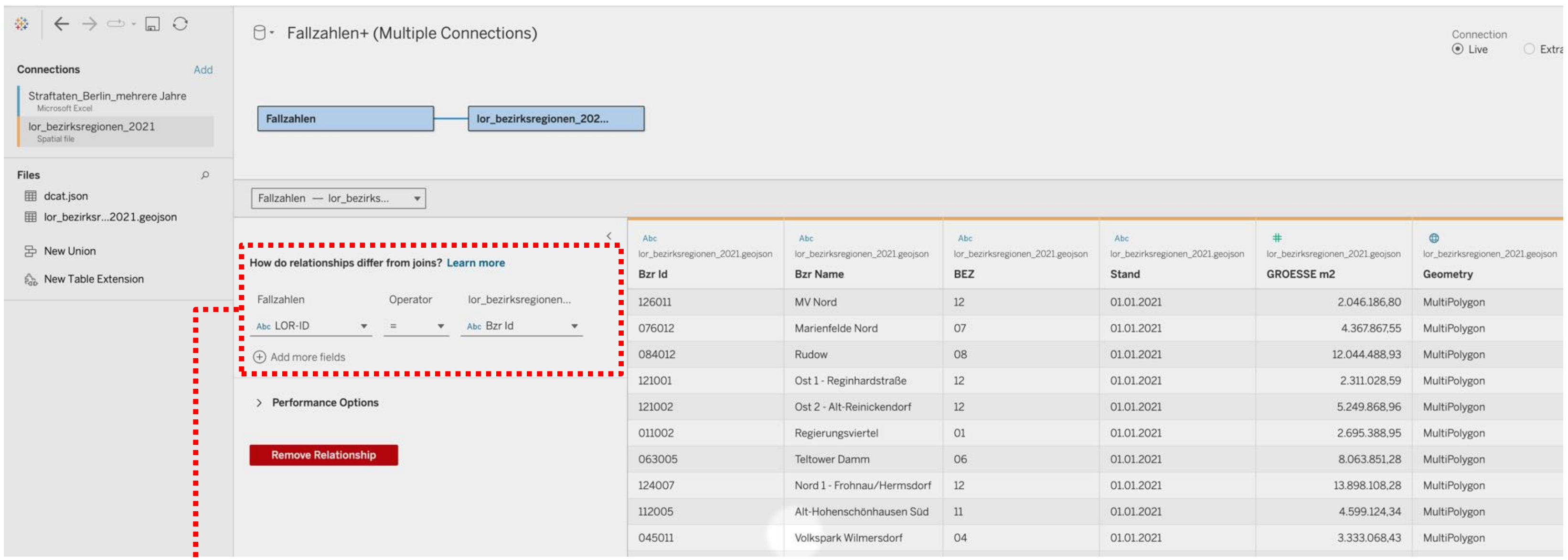
LOR files

- LOR all Levels.xlsx
- lor_bezirksregionen_2021.geojson: for LOR level BZR
- lor_planungsraeume_2021.geojson: for LOR level PLR
- bezirksgrenzen.geojson: for LOR level BEZ

Import Geo-spatial data and join with data table

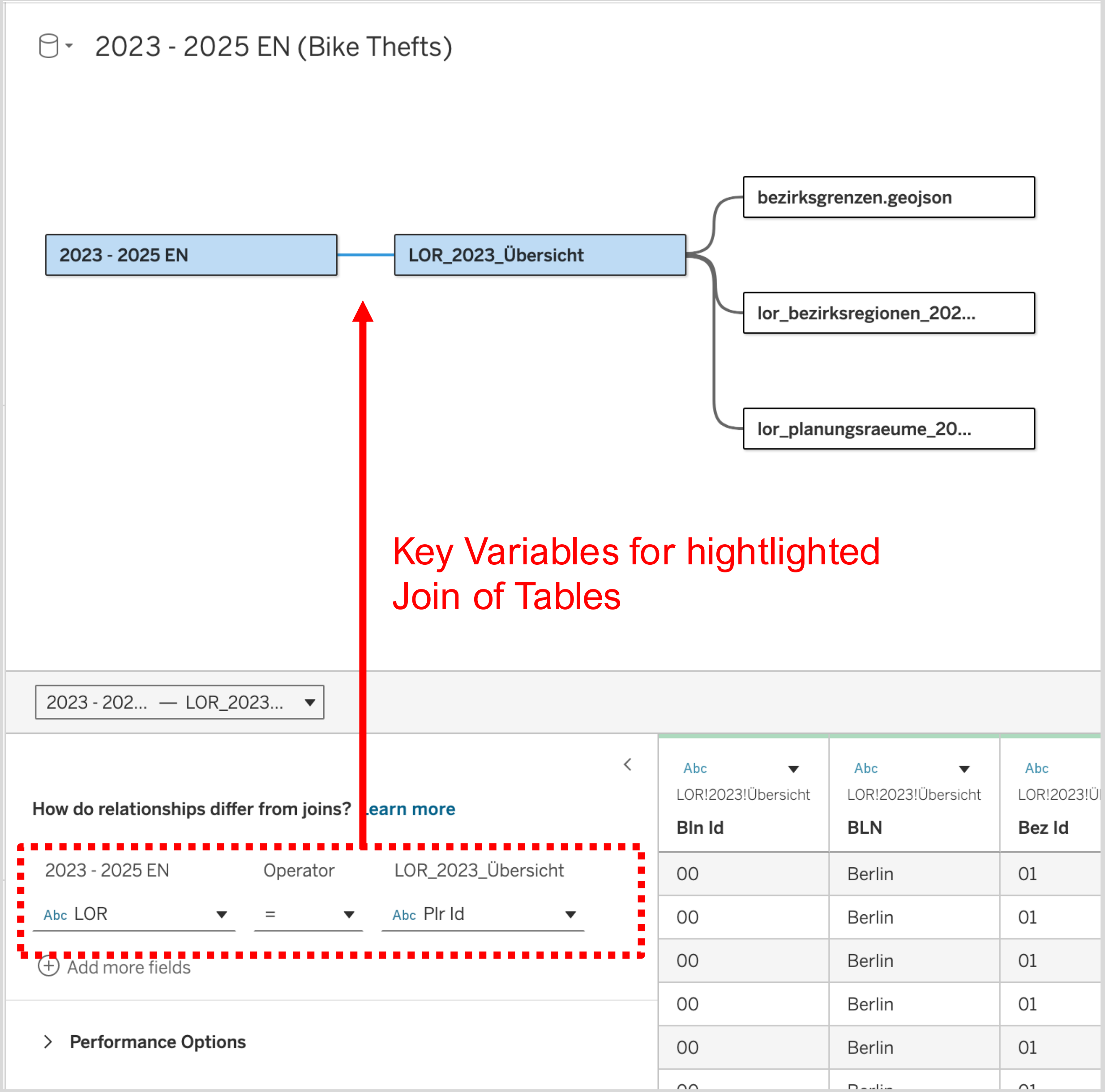
Import Geo data and join with data table

Here, multiple tables are linked together ("joined"). Unlike in SQL, R, or Python, Tableau does many things automatically – but not everything.

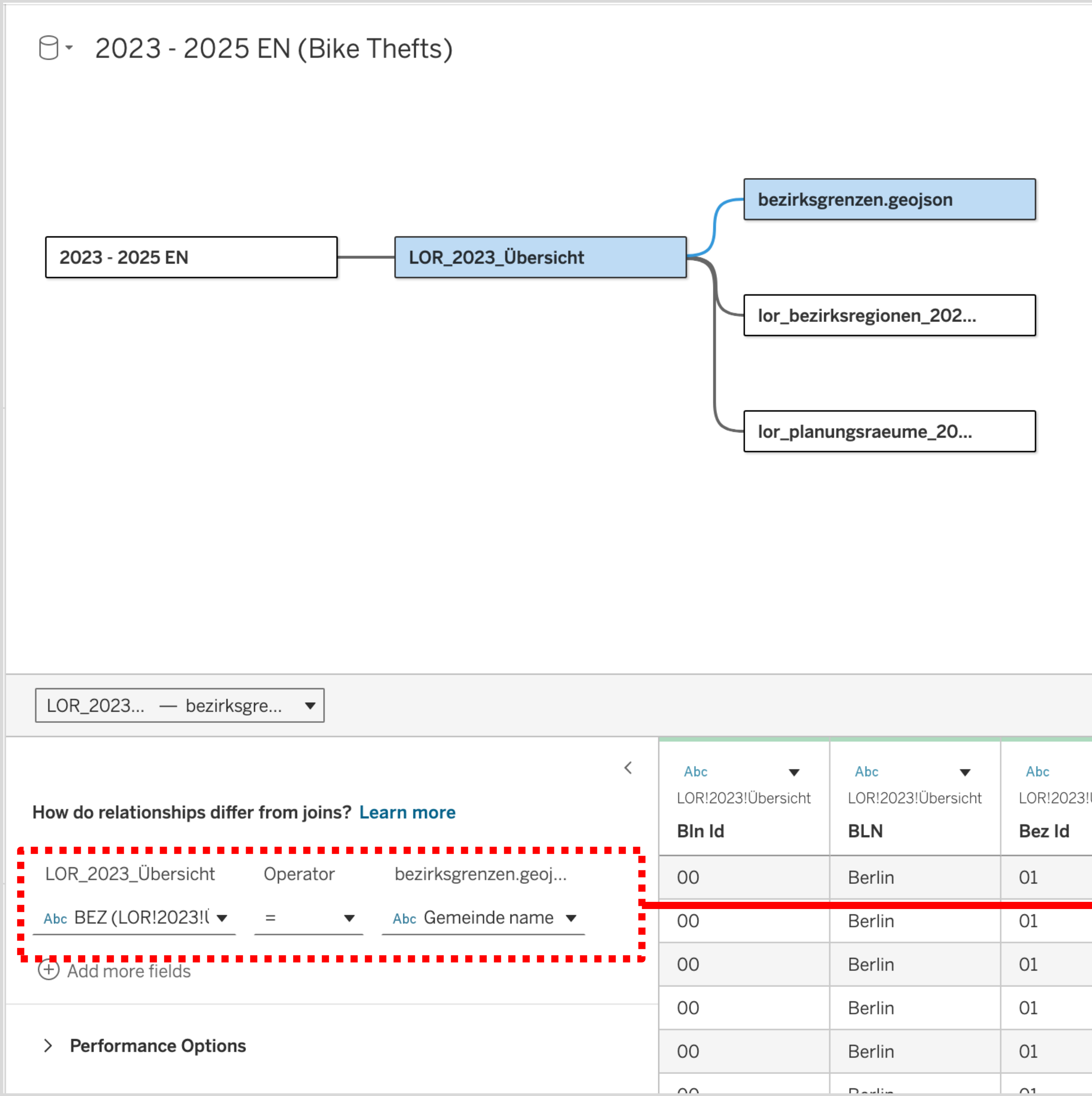


To perform joins, Tableau needs to know which columns in the different tables contain identical objects. Only then can Tableau merge the values from different rows ("map" them). In data science, this column is referred to as a "key variable"; in the context of SQL, it is also called a "join key" or "primary key" when it serves as a unique identifier in a table (as in our case).

Joins and Identifier Key Variables #1

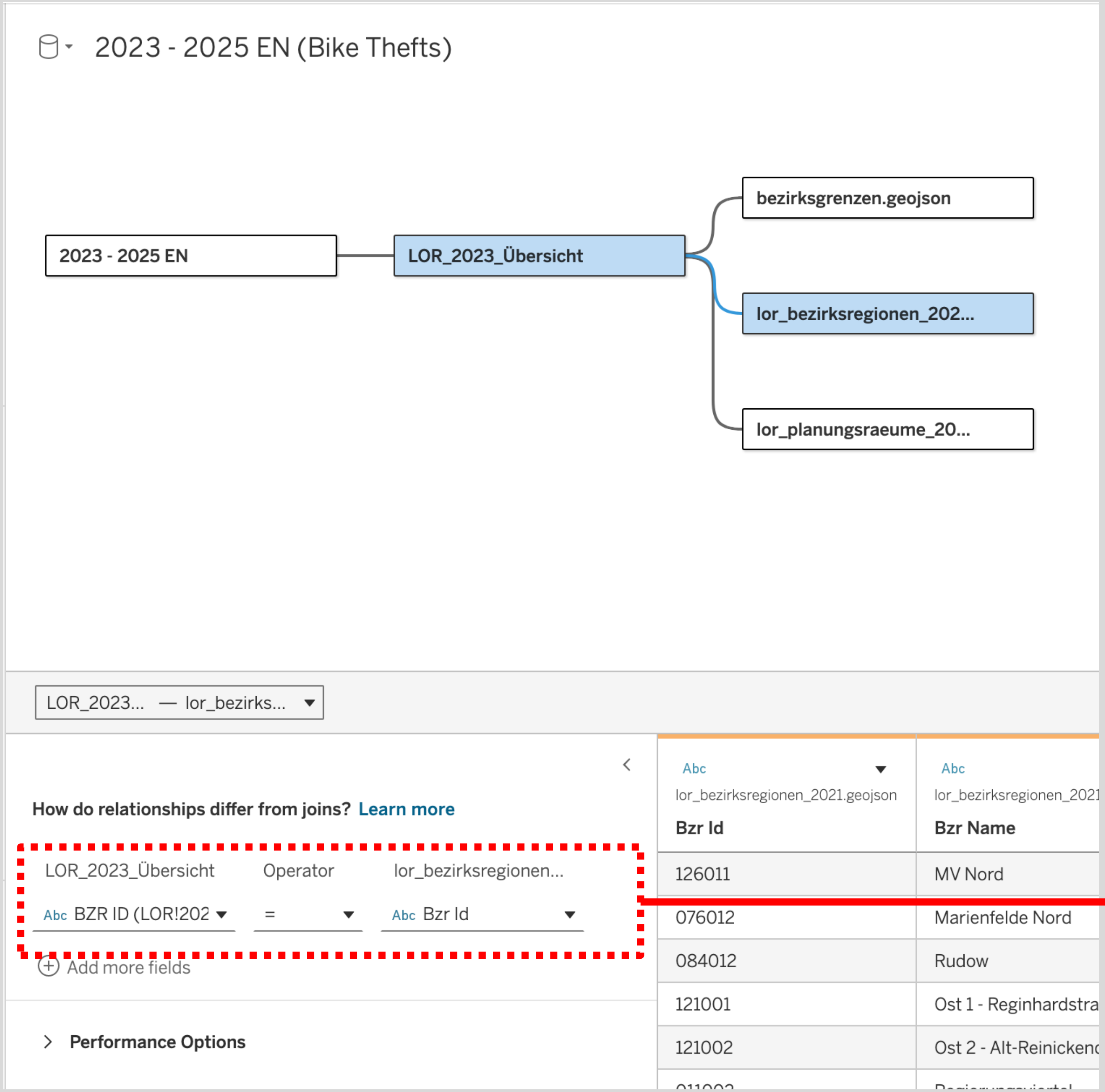


Joins and Identifier Key Variables #2



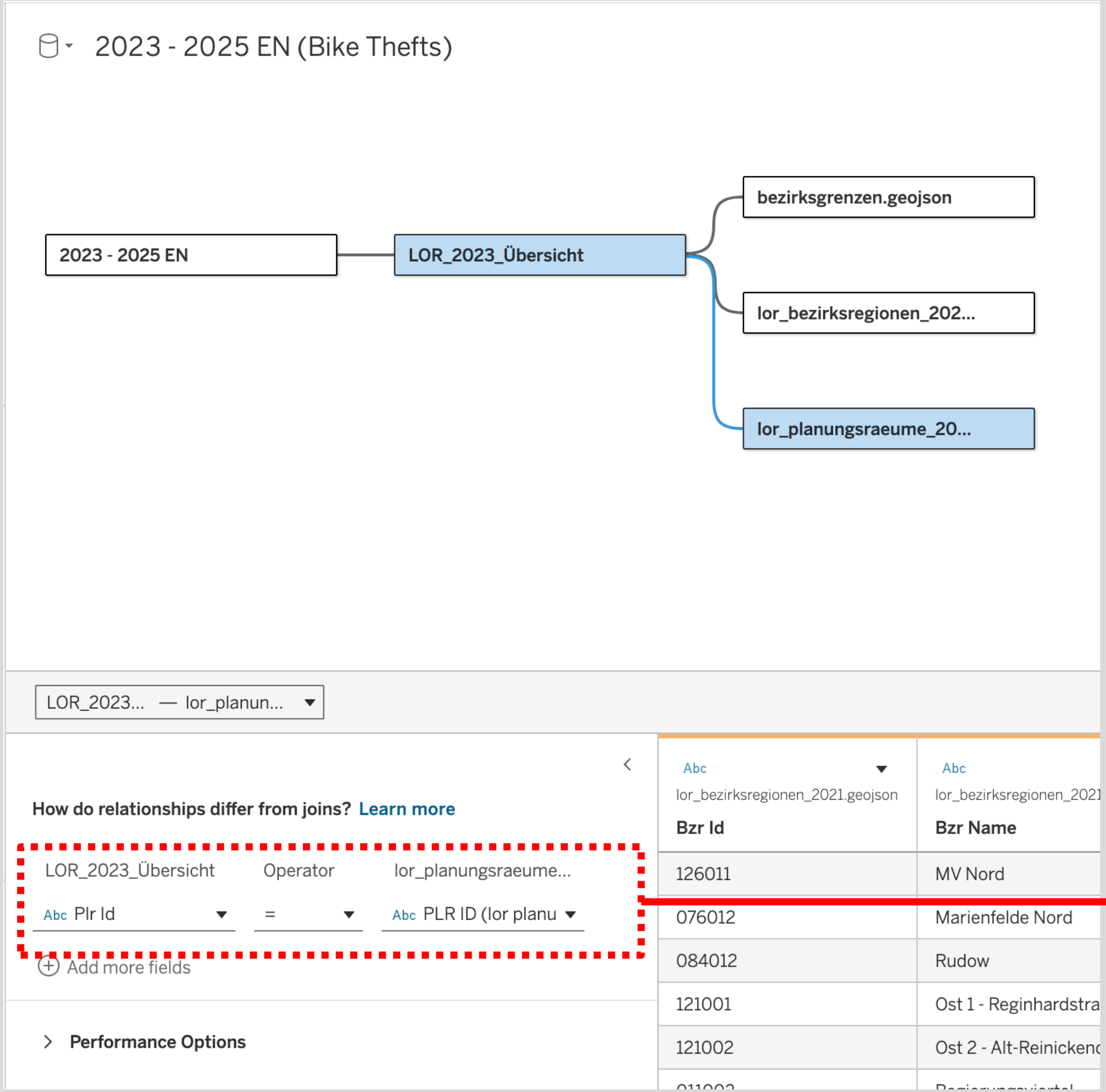
Key Variables for hightlighted
Join of Tables

Joins and Identifier Key Variables #2



Key Variables for hightlighted
Join of Tables

Joins and Identifier Key Variables #2



Key Variables for hightlighted Join of Tables